VidiVerify

The 1-click check for your videos

User Manual

Desktop application for video file integrity checking

Version 1.4.9

As of: July 2026

FREE, PRO and LIFETIME editions | For end users and technical professionals

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1. Overview and purpose

1.1 What is VidiVerify?

VidiVerify is a professional desktop application for the systematic integrity checking of video files. The software checks media files for errors, damage and completeness before they are archived, passed on, or put to productive use. Like the technical inspection a vehicle gets, VidiVerify performs the same role for digital video files: a structured, reproducible and documentable quality check.

The application is aimed both at private users who want to check media files from time to time, and at professionals in archiving, broadcast, post-production, video forensics, education and IT administration who need a reliable tool for daily use.

1.2 Typical problems

Digital video files can be damaged in many ways without this being noticeable immediately during playback. Typical causes of hidden errors include:

- Aborted uploads or interrupted downloads
- Errors when copying over unstable networks or damaged storage media
- Incomplete exports from editing or conversion software
- Loss of individual packets during streaming or recording
- Gradual corruption from the age-related decay of storage media
- Header structures that don't match the actual file content

> **[i] Why a dedicated check?**
> A file that opens and partially plays is not automatically error-free. Errors can be hidden in the last tenth of the file, individual frames can be damaged, or audio and video streams can run out of sync. Only a complete technical check reliably uncovers such problems.

1.3 Core idea of the application

VidiVerify bundles several established, high-performance and specialized plug-ins under a single, clearly structured user interface. These include FFmpeg and FFprobe for decoding and metadata analysis, MediaInfo for detailed media information, and Magika for content-based file type detection; for the more advanced image and frame analyses, the VidiVerify Runtime provides the bundled professional libraries PyAV and OpenCV. Each of these plug-ins makes its own contribution, and together they carry more weight than any single tool could deliver on its own. The application handles the elaborate parameterization automatically and presents the results in a clear, consistent form.

This makes professional file checking possible even without in-depth knowledge of the underlying command-line tools. The proprietary MAP module (Media Analysis Protocol) brings all check results together in a user-friendly display with clear color codes.

1.4 A concrete example: what VidiVerify takes off your hands

To illustrate the practical benefit, the following example shows the effort involved in checking a video file without VidiVerify, using the underlying command-line tools directly. A typical, complete check would have to call several plug-ins in sequence and then merge their outputs manually — something like this:

```
ffmpeg -hide_banner -v error -stats -xerror \
       -err_detect +explode+crccheck+bitstream+buffer \
       -fflags +discardcorrupt+genpts \
       -threads 4 -i "SampleVideo.mkv" \
       -map 0:v? -map 0:a? -map 0:s? \
       -c copy -f null - 2> check_log.log

ffprobe -v error -show_format -show_streams \
        -show_packets -show_frames \
        -of json "SampleVideo.mkv" > metadata.json

mediainfo --Output=XML "SampleVideo.mkv" > mediainfo.xml
```

This block of commands has to be typed by hand, adapted for each format, given the right paths, and then interpreted. The resulting log files run to thousands of lines of raw information that still need to be evaluated. Error messages appear in technical English, individual findings aren't automatically sorted by relevance, and the result is plain text. In practice, a solid assessment needs further plug-ins on top — MediaInfo for deeper container data, Magika for content-based type checking — each with its own calls, parameters and output formats.

VidiVerify takes care of all these steps automatically: the file is added by drag and drop, the check starts at the click of a button, the results from the plug-in ensemble appear in the MAP with color-coded ratings, and can be exported directly as a protocol. Several minutes of manual command-line work become a single click.

1.5 Three check levels

The application offers three check levels that build on one another, differing in depth and runtime:

Check level	Runtime	Purpose
Precheck	A few seconds	Checks accessibility, type and metadata of the file automatically on loading
Integrity check	30 seconds (sample)	Decodes a defined portion of the file and detects typical structural errors
Full check	From minutes to hours	Decodes the entire file completely and also uncovers hidden errors

2. Main features at a glance

VidiVerify combines a range of features that together provide end-to-end quality control for video files. The following overview lists the key capabilities of the current version. Features available exclusively in the PRO edition are marked accordingly.

2.1 File analysis and metadata

- Automatic reading of container, video and audio streams
- Display of resolution, aspect ratio, codec, bitrate, sample rate and bit depth
- Detection of file type and format family from the header signature (magic bytes)
- Quality classification by standard classes, from mobile resolution to Ultra HD
- Compatibility rating in the Media Lab: playback likelihood against six target profiles (Universal, Streaming, Smart TV, Mobile, Archive, Broadcast), based purely on technical metadata
- Tools and advisors in the Media Lab: CodecLab, Bitrate EquivalenceLab and Video SizeLab

2.2 Check methods

- Precheck runs automatically when a file is loaded
- Integrity check with a 30-second sample
- Full check across the entire file
- Optional strict mode, where the very first decoding error aborts the check
- VVProgressiv™ — a distributed, adaptive check that spreads short sequences across the entire runtime and automatically digs deeper wherever something looks suspicious PRO

With VVProgressiv™, the PRO edition takes the proven check mechanics to a whole new level: instead of only looking at the start of the file (integrity check) or the entire file in one pass (full check), VVProgressiv™ places distributed check windows across the whole runtime and escalates check depth precisely where anomalies appear. The result is a solid statement about the entire file in a fraction of the time a classic full check would take — VidiVerify's sequential, escalate-as-needed approach becomes intelligent and self-steering, taking even the full check itself to a new level.

2.3 Protocol creation and export

- The proprietary MAP (Media Analysis Protocol) with clear color coding
- Export of the MAP as a designed, single-page A4 PDF archive sheet (vector PDF)
- Dedicated "Diagnostic deep analysis" section in the protocol once the Diagnostic tool has been engaged
- MAP+ as a high-quality PDF archiving document with a professional layout PRO
- Export of an NFO file for media management systems such as Kodi, Jellyfin and Emby PRO

- File register with structured media and quality information
- Preview window with zoom and color highlighting

2.4 Robustness and diagnostics

- Diagnostic tool with three-tier deep analysis (streamcopy test, ffprobe deep analysis, decode layer) for stubborn cases
- Automatic escalation offer: if the standard check reports an anomaly, VidiVerify asks directly "Start diagnostics?"
- AI-assisted type integrity via the Google Magika plug-in (Apache-2.0, content-based, independent of extension and header)
- Media detail windows for audio and video with level, spectrum and bitrate curves, waveform display, and detection of dropouts and transcodes
- Read-only protocols, exception errors and runtime events
- Separate log files for subprocesses, exceptions, hangs and runtime
- Watchdog for the user interface
- Stable, anonymized device identifier for support packages
- Automatic assembly of a support package including relevant system information, optionally with a privacy scrub (paths, hostnames, IP/MAC masked)

2.5 Usability

- Quick-Check by right-click in Windows Explorer: a lightweight worker component delivers container structure, media data and an initial integrity finding within seconds, in a compact notice window — the key data becomes roughly ten times faster to grasp than reading native MediaInfo raw output; optionally hands off the finished result to the full application
- Context-menu integration in Windows Explorer: a top-level entry in the Windows 11 main menu as well as "Open with..." → VidiVerify — true to the 1-click promise, the app checks the file directly from Explorer, with no detour through file selection
- Clean Overview tab: three check steps shown as equal cards with pulsing status orbs, plain-language findings and a recommended action
- Photo-style preview frame with soft crossfade (slideshow) and automatic detection of cover art, snapshot or image series
- Live telemetry directly beneath the controls: system CPU, check-process CPU, RAM, network and read rate
- Honest progress bar with adaptive wall-clock estimation — no more jumps or stalls
- Three interface designs: Design mode, Windows classic, PLEX mode
- Multilingual interface in German, English, Chinese and Spanish — automatic detection of the system language on first launch
- Three CPU performance modes: gentle, balanced, high-performance

- Drag-and-drop support for video and audio files
- Protocol tab in the main window with detailed runtime information on the internal process

2.6 Updates and maintenance

- Independent update check for the application itself
- Separate update check for the integrated plug-ins
- Built-in download and installation wizard for plug-ins — cancellable, retryable and skippable
- Automatic switch to a VidiVerify mirror whenever an official plug-in source is unreachable or noticeably slow
- Archive-integrity check via SHA-256 hash values

2.7 Planned additional features (preview)

The following features are in development and will become available in upcoming versions. A detailed description can be found in chapter 14.

- AI forensics: an evidence-based assessment of technical AI indicators in video files, strictly separated into an AI-likelihood score and a confidence score, running entirely locally with no cloud upload PRO
- VVProgressiv™: a distributed, adaptive check of short sequences across the entire runtime that automatically digs deeper wherever something looks suspicious PRO
- Media Repair: rule-based repair suggestions for faulty files, always applied to a copy PRO
- Batch processing of multiple files in a single run, with a live results analysis PRO
- Statistics module with a learning, system-specific runtime forecast PRO
- Perceptual quality analysis (VQA) for visual quality assessment PRO
- YTUP™: a YouTube upload forecast with a quality prediction after re-encoding PRO
- "DVD check" tab for video and data DVDs as well as Blu-ray structures
- Porting and refactoring for macOS
- Further languages: Hindi, Russian, Portuguese

3. License model: FREE, PRO and LIFETIME

VidiVerify follows a transparent license model with three clear tiers: FREE, PRO and LIFETIME. FREE is, and remains, permanently free of charge for private users — a full-featured tool, not a demo. PRO unlocks the complete feature set and is mandatory for commercial use. LIFETIME is PRO in its most unrestricted form: pay once, keep everything, forever.

3.1 FREE — the full-featured starting point

FREE contains all the core functions for reliable video-file integrity checking and is usable without restriction — for personal use, for learning, and for any non-commercial purpose. FREE is deliberately not a stripped-down trial mode, but a complete, everyday-ready checking tool.

- Precheck, integrity check and full check
- MAP (Media Analysis Protocol) with color coding
- Export of the MAP as a designed PDF archive sheet
- Media Lab: compatibility rating against six target profiles, plus the CodecLab, Bitrate EquivalenceLab and Video SizeLab tools
- Media detail windows for audio and video with dropout and transcode detection
- All display designs and all four languages (German, English, Chinese, Spanish)
- All performance and configuration settings
- Plug-in management and update checking

FREE can optionally be registered with a license number. VidiVerify already shows the current license status on the splash screen at startup.

3.2 PRO — VidiVerify without limits

PRO is VidiVerify at its fullest extent. A PRO license is aimed at everyone who wants more: at commercial users, for whom PRO is mandatory under our terms of use; at organizations and archives with demanding requirements; and at ambitious private users who simply want the complete tool in their hands. PRO includes the entire FREE feature set and, on top of that, unlocks the powerful professional modules. Once licensed, everything is unlocked automatically — no reinstall, no detours.

> **[PRO] What PRO unlocks**
>
> All PRO features in this manual are marked with a yellow badge. Example:
>
> Media Batch **PRO**
>
> **The PRO expansion of VidiVerify:**
>
> - MAP+ as a high-quality, archive-ready PDF
> - NFO export for Kodi, Jellyfin and Emby

- Media Repair — lossless repair of typical error patterns
- Media Batch — batch processing of entire folders
- AI forensics, VVProgressiv™, statistics module, perceptual quality analysis and YTUP™

The PRO expansion keeps growing with every update — start today, and grow with it.

A PRO license applies to one version generation at a time — and that is a deliberate choice. Every generation brings a noticeable leap in capability, and your PRO license carries that entire leap with it, from the very first new feature to the very last. This keeps you flexible: you decide afresh with every generation — and you get a price-to-performance ratio that is hard to match.

3.3 LIFETIME — pay once, keep everything, forever

LIFETIME is the greatest freedom in the model: once purchased, PRO stays active permanently — no subscription, no recurring cost, no expiry date. Where a PRO license applies to one version generation at a time, LIFETIME removes that boundary entirely: every future generation leap is already included, every capability expansion to come is part of the deal. For anyone using VidiVerify over the long term, LIFETIME is the smartest choice — securing the full future feature set at today's price of entry.

3.4 Automatic unlocking

Once licensed, the application recognizes the valid license automatically at startup and activates all PRO features without any further manual steps. No reinstallation is required. The license check runs locally — a permanent internet connection is not required for operation.

3.5 Switching editions and data retention

You can switch from FREE to PRO or LIFETIME at any time by purchasing a license. All existing settings, protocols and check results are fully retained. Likewise, if a time-limited license expires, no data is lost — the application simply reverts to the FREE feature set.

4. Benefits for users and organizations

VidiVerify is deliberately designed as a tool that delivers compelling value both for single-user operation and for organizational deployment. The following sections highlight the key benefits.

4.1 For end users

[+] Save time and avoid risk

Instead of reviewing video files by hand or calling up several separate tools, VidiVerify runs all the relevant checks in a single pass. Errors are caught before a damaged file gets archived or passed on.

- A clear, guided workflow with no command line
- Meaningful, easy-to-read check protocols with color coding
- Instantly visible status indicators instead of cryptic error messages
- Three interface designs for different preferences
- Configurable CPU load, so the system stays responsive during a check

4.2 For technical professionals

- Full control over time budget, strict mode and CPU settings
- Detailed event logs in the Protocol tab for diagnostics and traceability
- Clean separation between the program directory and user data
- Freely accessible log files in plain-text format
- Stable, reusable check logic as a foundation for reproducible results
- Automatic assembly of a support package for submission to the support team

4.3 For organizations and archives

[*] Quality assurance as a documented process

Every check can be archived as a PDF. This creates an audit-proof protocol that can be used for quality assurance, incoming-goods inspection or forensic documentation. The PRO edition additionally provides MAP+ as a high-quality archiving PDF.

- Audit-proof logging of check results
- Consistent check criteria across teams and locations
- Open, traceable storage of settings and protocols
- No cloud dependency: checking happens entirely locally
- Stable device identifier so the support team can identify your device across cases
- NFO export for seamless integration into media management systems PRO

4.4 Economic benefits

Aspect	Benefit
Less investigative effort	Faulty files are caught early, before time is invested in downstream steps
Avoiding data loss	Early detection of damaged archives saves later recovery costs
Fewer manual checks	Automated check runs replace spot-checking individual files by hand
A consistent check standard	Every checker uses the same criteria and protocol structures
A clear license model	A free FREE edition for personal use, a dependable PRO edition for professional deployment

5. Installation and setup

5.1 System requirements

Component	Recommendation
Operating system	Windows 10 or Windows 11 (64-bit), macOS planned
Memory	At least 4 GB, 8 GB recommended
Disk space	At least 200 MB for the application, plus space for plug-ins
Processor	64-bit processor with at least two cores
Screen resolution	At least 1105 × 753 pixels
Permissions	Standard user rights are sufficient; administrator rights are only needed for the initial installation

5.2 Installation steps

1. Download and run the installer.
2. Confirm the target folder (typically C:\Program Files\VidiVerify).
3. Run the installation and start the application.
4. On first launch, the application automatically runs a plug-in check.
5. If any plug-ins are missing, a guided download wizard is offered.

> **[+] Simple and safe**
> The installation wizard guides you through the whole process. The required plug-ins can be downloaded right away, or loaded later from within the application.

5.3 First launch and plug-in check

On first launch, a splash screen appears and remains visible briefly. During this time, the application checks the availability of all required plug-ins. If any plug-ins are missing, a dialog for installing them opens automatically. You can either start the download immediately or skip it, in which case individual check features only become available once the plug-ins have been added.

> **[i] Fresh-install resilience: fallback source for plug-ins**
> If a plug-in's official source is unreachable at download time (for example due to rate limiting or a brief outage), VidiVerify automatically falls back, per plug-in, to a self-hosted mirror at vidiverify.de, and verifies the

> archive against a fixed SHA-256 value. Only the plug-in whose source failed is affected; all others still load from the official source as usual. The later update check pulls the newer official version regardless.

5.4 Language selection

The application currently supports four languages, available from the Settings dialog.

- German (the default on German-language systems)
- English
- Chinese (中文)
- Spanish (Español)

Hindi, Russian and Portuguese are also planned.

5.5 Licensing and activation

Licensing is handled through the License module in the Settings dialog. Even FREE can optionally be registered there with a license number. Switching to PRO or LIFETIME is done by entering the relevant license key — the full feature set is then unlocked automatically. The check runs locally and remains valid without an internet connection. VidiVerify already shows the current license status on the splash screen at startup.

5.6 Uninstallation

Uninstallation is performed through the bundled uninstaller (unins000.exe) in the program folder, or via the Windows Control Panel. During interactive uninstallation, VidiVerify asks two yes/no questions, each defaulting to "No" so that nothing is deleted by accident:

- Should the machine-wide plug-ins under %ProgramData%\VidiVerify be removed? (Choosing "No" keeps FFmpeg, MediaInfo and Magika in place, so they don't need to be downloaded again on a fresh install.)
- Should your personal VidiVerify data in the user profile (%LocalAppData%\VidiVerify) also be removed? (Choosing "No" keeps settings, the preview cache, protocols and the Quick-Check inbox in place, and they will be picked up again on a later reinstall.)

This lets you either remove VidiVerify completely, or simply replace the program without losing configuration and plug-ins.

6. Use, operation and workflows

6.1 Start screen and main window

After launch, the application's main window opens. It consists of several areas that mirror the typical workflow, from file selection through to the check protocol.

- File selection with a Browse button and a drag-and-drop area
- Overview tab as the entry point: three check steps shown as equal cards with pulsing status orbs, plain-language findings and a recommended action
- Photo-style preview frame with a light matte border — cover art, snapshot or image series are detected automatically; for Dolby Vision material, a note points out that colors may look different on players without Dolby Vision support
- Information field with container, duration, and video and audio properties
- Check-level selector between integrity check and full check
- Live telemetry beneath the controls: system CPU, check-process CPU, RAM, network, read rate
- Progress display with elapsed time and status (adaptive wall-clock estimation)
- Results area with the classified check result (MAP)
- Media Lab tab: compatibility rating of the loaded file, plus tools and advisors (CodecLab, Bitrate EquivalenceLab, Video SizeLab) — see chapter 7
- Protocol tab with detailed runtime information on the internal process

6.2 Typical workflow

1. Start the application.
2. Select a video file, add it by drag and drop, or hand it over via right-click in Explorer ("Open with..." → VidiVerify).
3. Wait for the automatic precheck (a few seconds).
4. Choose a check level: integrity check or full check.
5. Start the check.
6. Watch the progress, or cancel if needed.
7. Review the result in the MAP (color-coded display).
8. If needed, look at the details in the Protocol tab.
9. If anomalies show up, engage the Diagnostic tool (see 6.9).
10. Save the MAP as a PDF archive sheet, or (PRO) as MAP+.

6.3 Precheck

The precheck starts automatically as soon as a file has been selected. It checks basic accessibility and reads the metadata using the FFprobe plug-in. The result appears immediately in the information display. This level delivers an initial assessment within a few seconds, without fully decoding the file.

> **[i] A note on network paths**
> Files on network paths may need a longer precheck, depending on connection quality. The network-source policy setting lets you specify whether such sources should be ignored, flagged with a warning, or blocked.

Even at the precheck stage, VidiVerify flags known playback stumbling blocks that are deliberately not treated as integrity errors. In these cases the file is technically sound — the note only concerns playback compatibility on certain players:

- Fragmented streaming MP4 without an index (typical for IP-camera and NVR recordings): some players can't use duration and seeking correctly. The precheck card turns yellow with a short note; the finding also feeds into the diagnostics as an anomaly, with a recommendation to re-wrap the file once (remux).
- HEVC/H.265: a neutral note with no traffic-light effect, stating that some players need an additional HEVC video extension for this codec format.

> **[i] Yellow doesn't mean "broken" here**
> A compatibility note turns the precheck card yellow because something was detected and explained — not because the file is damaged. The actual check results (integrity, full and diagnostic checks) are unaffected by this.

6.4 Integrity check

The integrity check decodes the first 30 seconds of the file. This is typically where the container's most important header structures and index information live. Errors in this area point to structural defects and are detected reliably. The runtime stays manageable, since only a defined portion is processed.

6.5 Full check

The full check decodes the entire file completely. This is the only way to catch errors located further into the file, or errors affecting only individual frames. Runtime depends on file size, codec and available processing power, and can run to several hours for long recordings.

6.6 Progress and cancellation

During a check, the progress display shows the current percentage as well as the elapsed time. A running check can be stopped at any time via the Cancel button. The application asks for confirmation first, to prevent accidental cancellations.

6.7 The Protocol tab

Alongside the check result, the main window also provides the Protocol tab. It shows detailed runtime information about the internal process: processes started, timestamps, plug-in messages, detected stream information and technical notes. This tab is especially useful for technical professionals who want to trace the exact course of a check or make sense of anomalies.

The protocol is laid out as a two-column view: on the left, "Latest action" gives an at-a-glance summary of the most recent, currently running, or just-completed activity; on the right, "Running log" shows the complete, chronological chain of events. This keeps the current step visible at all times, without losing sight of the overall course of events. Both columns use a zebra-striped layout.

6.8 Check protocol and file register

Once a check finishes, VidiVerify automatically creates the MAP (Media Analysis Protocol). It is saved as a designed, single-page A4 PDF archive sheet. If the Diagnostic tool was engaged, a "Diagnostic deep analysis" section also appears, with an overall rating and findings. The PRO edition additionally provides MAP+, an even more extensive archiving PDF for professional filing purposes.

6.9 Diagnostic tool

The Diagnostic tool is the answer-generating deep analysis for stubborn cases. It comes into play when the standard check reports an "anomaly" and you need a clear-cut cause. Three specialized check layers work together:

1. Streamcopy test: checks whether the container can be re-wrapped cleanly (a fast structural sanity test), uncovering timestamp and packet-order problems.
2. ffprobe deep analysis: an in-depth read of packet and frame information, including timestamp, index and stream structures.
3. Decode layer: trial decoding of targeted frame ranges, uncovering codec-specific defects that aren't visible at the container level.

From these three layers, VidiVerify derives a clear diagnosis — with a cause, a severity level, and a concrete recommendation for action. It reliably detects tape errors from VHS digitization, faulty DVD rips, aborted downloads, tampered streams and damaged containers.

> **[*] Understands quirks instead of raising false alarms**
> The Diagnostic tool recognizes DVD cell transitions and H.264/HEVC B-frame structures for what they are: normal encoder behavior, not a defect. No more confusing red flags on perfectly sound files.

Automatic escalation offer

If the standard check reports something anomalous, VidiVerify asks directly: "Start diagnostics?". One click, and the tool delivers. The check protocol then also gains a "Diagnostic deep analysis" section with an overall rating and findings.

6.10 Quick-Check in Windows Explorer

Quick-Check is the fastest way to check a video file — without opening the full application at all. A right-click on a video file in Windows Explorer is enough: a lightweight, standalone worker component reads the container structure, the key media data and an initial integrity finding within seconds, and shows them in a compact notice window. This makes the relevant core data noticeably faster to take in than reviewing native, technical raw output.

- Launched directly from the Windows 11 main context menu (a top-level entry with its own icon)
- The notice window shows status (OK, Warning, Error), a short finding, and the key media data — several windows can be open at the same time
- Optionally hands off to the full application: the already-computed result is carried over, and the file is loaded there immediately
- Multilingual in German, English, Chinese and Spanish

> **[i] Also available: "Open with..." integration**
> Independently of Quick-Check, VidiVerify can also be launched via "right-click → Open with... → VidiVerify" — the full application then opens with the file already loaded and immediately starts the precheck. VidiVerify deliberately does NOT become the default player this way: double-clicking a video file behaves exactly as before, and VidiVerify simply appears as an additional option in the list.

6.11 Media Test — detail windows for audio and video

From the Media Analysis tab, two detail windows can be opened for a loaded file that go far beyond a simple metadata display. Both are part of the FREE feature set, open as standalone windows, and follow the language of the main application. They become active as soon as a valid, prechecked file with usable data is loaded.

- Audio detail window: level history, statistics, bitrate curve, frequency spectrum and waveform — with detection of dropouts (brief audio interruptions) and transcodes (bitrates inflated after the fact). The charts can be zoomed and navigated much like in audio-editing software; where a file has several audio tracks, you choose which one to examine.

- Video detail window: technical structure and bitrate analysis with a high-performance preview. During analysis, a centered progress ring shows the current state; the window is also the precursor to the later repair feature (Media Repair, PRO).

The detail windows close automatically together with the main application, and whenever a new file is loaded. Before replacing a loaded file, VidiVerify asks for confirmation as a safeguard.

6.12 Checking pure audio files

As of version 1.4.9, VidiVerify treats pure audio files as a full, dedicated category — no longer as "video with no picture", and certainly not as a file that looks broken by mistake. When an audio file is loaded (Appendix C lists the supported formats), VidiVerify reliably recognizes it via magic bytes and the Magika plug-in, switches the interface to an audio-appropriate layout — cover art instead of a video preview, audio-specific key figures in the information field — and runs the same structured check levels as for video: precheck, integrity check and full check.

The audio check shows its full strength in the audio detail window (Media Analysis tab → Audio details). It makes the sound visible and uncovers problems you would otherwise only notice by listening closely:

- Dropout detection: brief audio interruptions, clicks and silent gaps are flagged visibly in the level and waveform display — ideal for catching recording, editing or transmission errors before they end up in the finished product.
- Uncovering inflated bitrates (upcoding/transcode): from a tell-tale frequency cutoff, VidiVerify detects when a file is labeled with a high bitrate but the actual audio material really comes from a lower-quality source — for example, an MP3 declared as 320 kbit/s that was in fact upsampled from a 128 kbit/s file.
- Visible audio analysis: level history, frequency spectrum, bitrate curve and waveform in one window, with zoomable, navigable charts much like in audio-editing software.

[*] Honest sound instead of a false label

For music and speech recordings, what matters is what's actually in the signal — not what the file header claims. VidiVerify checks both: the file's formal integrity AND the real quality of the audio. Inflated bitrates and hidden dropouts are exposed before they become a problem.

7. Media Lab

Media Lab is a dedicated tab in the main window and the launch pad for VidiVerify's more advanced, file-related tools — fully included in the FREE feature set. The tab is theme-aware and switches live with the selected interface design (Design mode, Windows classic, PLEX mode). It is organized into two areas: the compatibility analysis of the loaded file, and the tools and advisors.

7.1 Compatibility analysis

The compatibility analysis rates a file's technical playback likelihood against six typical target profiles — based purely on the harmonized metadata from FFprobe and MediaInfo. It deliberately makes NO judgement about picture quality; that is reserved for the later perceptual quality analysis. The core benefit: VidiVerify never makes an untenable claim like "plays everywhere" — instead, it derives a technically grounded assessment for each target from objective file characteristics.

Target profile	Reference point
Universal	Broad playability across as many devices and players as possible
Streaming	Delivery via the web and streaming services (e.g. fMP4 with an index)
Smart TV	Playback on common smart TVs and set-top boxes
Mobile	Playback on smartphones and tablets
Archive	Long-term, lossless preservation
Broadcast	Professional broadcast and production environments

Each profile receives a traffic-light rating with four levels:

- Green — high: the file reliably meets the profile's requirements
- Yellow — good, broadly supported: plays well, with minor limitations compared to the reference point
- Orange — limited: playback is likely only under certain conditions
- Red — low: technically unsuitable for this profile

[i] Conservative rather than generous

When characteristics conflict or are unclear, the rating deliberately errs on the side of caution. Special cases are recognized and named: Dolby Vision Profile 5 (without an HDR10 base layer) shifts the colors on players without Dolby Vision support, and is therefore disqualified for Universal, Streaming and Mobile; a fragmented MP4 without an index is a hard exclusion rule for Streaming. This keeps the assessment reliable, rather than flattering a sound file or overlooking a problematic one.

The result is displayed in a dedicated window with a profile rail and a double ring: the inner, thicker ring summarizes all the rules in the corresponding traffic-light color, while the outer, thinner ring represents just the hard must-pass rules. Explanatory tiles set the requirement for each characteristic against the actual value found, translated into plain language. The compatibility card in the Media Lab also shows a short traffic-light summary for the loaded file, and becomes active as soon as a valid, prechecked file is loaded. The window is multilingual (German, English, Chinese, Spanish).

7.2 Tools and advisors

The second area bundles three compact tools that answer technical questions about codec, bitrate and resolution without requiring deep expertise.

- CodecLab: classifies codecs and their properties, and helps choose the right format for a given purpose.
- Bitrate EquivalenceLab: translates bitrates between codecs — what bitrate in a more modern codec corresponds to a given bitrate in an older one, at comparable quality?
- Video SizeLab: estimates the effect of scaling. It distinguishes plain upscaling (UPS) from AI-assisted methods (VSR), and walks from a source to a target format, with a slider to try out different bitrates.

[PRO] The gateway to the professional tools

Media Lab is also the launch pad for the PRO tier: from here, the paths lead to the perceptual quality analysis and to lossless repair (Media Repair). These professional tools are part of the PRO expansion and will be added via updates — so Media Lab grows right along with it.

8. The proprietary MAP (Media Analysis Protocol)

8.1 Idea and purpose

The MAP (Media Analysis Protocol) is VidiVerify's proprietary centerpiece. It translates the complex technical results from deep, IT-level check runs into a clear display that any user can understand. The MAP's central strength lies in assessing complicated technical values from in-depth check runs and making them visible through simple color codes.

> **[*] The wow effect: technology meets clarity**
>
> Raw data from FFprobe, FFmpeg and MediaInfo often runs to thousands of values and parameters. The MAP distills this into statements you can grasp at a glance: color codes instead of columns of numbers, understandable classifications instead of raw technical values. This is what makes professional video analysis usable for everyone with VidiVerify.

8.2 What the MAP delivers

- A summary of all check levels in a single overview
- A color-coded display of the overall result (see Appendix B)
- Core technical values classified in plain language
- Notes on anomalies in understandable language
- A link between the raw protocol and the user-friendly rating
- Direct export as a designed, single-page A4 PDF archive sheet

8.3 Structure of the MAP

The MAP is rendered as a designed, single-page A4 portrait archive sheet. It is organized into clearly separated areas that together form a complete picture of the checked file on a single page:

Area	Content
Side bar	Narrow strip on the left with the "MAP" label and a subtitle
Header	Logo and "VidiVerify" wordmark, "MEDIA ANALYSIS PROTOCOL" centered, "ARCHIVE AND CHECK SHEET" on the right; the file title beneath
Identification	Year, genre and duration at a glance
Video / Audio	Two columns: key picture and sound values as key-value pairs, each with a color-coded verdict chip in the level's color
Precheck / Type integrity	Type result plus the Magika finding (label and score) as two compact status fields

Area	Content
Checks	Three equal-width panels for the integrity, full and diagnostic checks, each with a status orb, result and short text
Description	Understandable classification and anomalies in plain language
File	Size, path, created and modified dates
Verification & output	Export details, file identifier, profile and device identifier (VV-ID), with a closing line (version, note, page number)

[*] Multilingual, right down to the font

The MAP PDF embeds a bundled Unicode font, rendering not just German umlauts but also genuine Chinese, Cyrillic and Greek script cleanly — with no need for a matching system font to be installed. Thanks to font subsetting, the finished PDF still stays small.

8.4 MAP export

The MAP can be saved directly from the application as a PDF. The preview in the main window and the export use the same renderer, so the saved document matches the on-screen view exactly. The PDF is suitable for archiving, sending, and audit-proof documentation.

[PRO] MAP+ as an archiving PDF (PRO)

The PRO edition additionally provides MAP+: a high-quality PDF archiving document with a professional layout. It includes an expanded display and additional technical sections, and is optimized for long-term archiving, forensic documentation, and check protocols handed over to clients.

8.5 NFO export for media management (PRO)

In addition to the MAP export, the PRO edition can generate an .nfo file on request. This format is read by media management systems such as Kodi, Jellyfin and Emby, and automatically enriches the media library with technical information. The NFO file is generated directly from the existing analysis results and follows common standards for robust, long-term-maintainable integration.

9. The integrated plug-ins

VidiVerify relies on a lineup of proven, high-performance specialist plug-ins, each handling a clearly defined task. The application coordinates this ensemble of plug-ins, parameterizes the tools automatically, and brings their results together in the MAP. For the user, this creates a single, consistent interface, while proven tools work in the background whose combined value goes well beyond what any single plug-in could deliver.

9.1 FFmpeg

FFmpeg is a globally recognized plug-in for processing and decoding media files. Within VidiVerify, FFmpeg handles the actual decoding during the integrity and full checks. The application calls FFmpeg with finely tuned parameters, so that errors are reliably detected and logged.

- Core task: decoding and checking all streams
- Strengths: broad format support, very robust error detection
- Used in: integrity check and full check

9.2 FFprobe

FFprobe is FFmpeg's sister plug-in, used for reading metadata and stream information. VidiVerify uses FFprobe in the precheck, and to populate the MAP with basic technical data.

- Core task: reading containers, streams and codec information
- Strengths: fast analysis without full decoding
- Used in: precheck and metadata enrichment of the MAP

9.3 MediaInfo

MediaInfo is a specialized plug-in for the detailed analysis of media files. It provides especially thorough and precise information on containers, codecs, streams and extended parameters. VidiVerify uses MediaInfo as a supplementary source and to cross-check results. Since v1.4.7, MediaInfo also detects Dolby Atmos and DTS:X structures in main audio tracks. Main-track selection additionally takes locale and the default tag into account — a German-language app finds the German audio track even if the container's default flag points elsewhere.

- Core task: in-depth metadata analysis with a broad range of parameters
- Strengths: very thorough technical information, a complementary perspective, Atmos and DTS:X detection
- Used in: supplementary analysis, foundation for the compatibility rating

9.4 Magika

Magika is a plug-in published by Google for content-based file type detection. According to Google, it is used in production in Google's own security and file pipelines. Magika has been a permanent part of the application since v1.4.7. While the classic magic-byte check reads the file header, Magika works on the file content using a model — both methods are shown side by side in the Overview tab. What one method misses, the other often catches, so the two together give a reliable verdict. Magika is licensed under Apache-2.0 and weighs in at around 10 MB.

For swapped or disguised files (for example a PDF with a .mp4 extension, or an EXE inside a video container), VidiVerify clearly marks the result as "not a media file", with a concrete note on how to correct it.

- Core task: content-based file type detection
- Strengths: robust detection even for renamed or manipulated files
- Used in: a second opinion alongside the classic magic-byte check, visible in the Overview tab

9.5 VidiVerify Runtime (PyAV and OpenCV)

The advanced Media Lab and Media Test windows run in their own, proprietary runtime environment — the VidiVerify Runtime, built on Qt. It is not a separately downloadable command-line plug-in, but a bundled package that is loaded automatically the first time it's needed — primarily from VidiVerify's own domain (dl.vidiverify.com), with a mirror as a fallback. The download shows named progress and can be cancelled, retried or skipped; the runtime is visible in the Settings dialog and can be loaded from there on demand.

The runtime bundles high-quality Python media libraries: PyAV binds the full FFmpeg library for frame-accurate decoding, and OpenCV — the industry standard for computer vision — provides the foundation for the image and frame analyses in the Media Test detail windows. For the future perceptual quality analysis (VQA, PRO, see chapter 14), OpenCV will also provide the foundation.

- Core task: graphical detail and tool windows, plus image and frame analysis
- Includes: PyAV (FFmpeg binding), OpenCV (computer vision) and the Qt interface
- Delivery: automatic on-demand loading with named progress

9.6 How the plug-ins work together

Each plug-in handles its own specialized task and is orchestrated by VidiVerify. The results flow centrally into the MAP, where they are prepared and color-rated for the user. This architecture results in a reliable, modular and future-proof platform.

10. Configuration and parameters

The application's settings are managed in the Settings dialog, which opens from the menu or the Settings button in the main window. All changes are saved to the vidiverify_settings.json file in the user folder.

10.1 Plug-in paths

The plug-ins reside machine-wide under %ProgramData%\VidiVerify\tools — shared by every user of the machine, and outside any individual user profile. The following defaults refer to this path.

Setting	Description	Default
FFmpeg path	Path to the FFmpeg executable	%ProgramData%\VidiVerify\tools\ffmpeg\ffmpeg.exe
FFprobe path	Path to the FFprobe executable	%ProgramData%\VidiVerify\tools\ffmpeg\ffprobe.exe
MediaInfo path	Path to the MediaInfo plug-in	%ProgramData%\VidiVerify\tools\mediainfo\
Magika path	Path to the Magika plug-in for DeepCheck	%ProgramData%\VidiVerify\tools\magika\
VidiVerify Runtime	Qt runtime environment for Media Lab and Media Test — visible in the Settings dialog and loadable from there	%ProgramData%\VidiVerify\tools\vv_runtime\

10.2 Check behavior

Setting	Description	Default
Strict mode	The very first decoding error aborts the check	On
Time budget	Time limit per check in seconds, 0 means no limit	0 (unlimited)
Maximum log lines	Maximum number of lines in the runtime log	10,000

10.3 Performance settings

CPU usage can be adjusted via three predefined performance modes, offering a sensible balance between check speed and system responsiveness.

Performance mode	Processor cores	Priority	Typical use
Gentle	One core	Low	Checking in the background during normal work
Balanced	Half the cores	Normal	Default setting for most cases

Performance mode	Processor cores	Priority	Typical use
High-performance	All cores	High	Fastest possible check of individual files

10.4 Display and language

Setting	Options
Display language	German, English, 中文, Español (more planned)
Interface design	Design mode, Windows classic, PLEX mode
Compact mode	Automatic, On, Off — a space-saving layout for small laptop resolutions (e.g. 1280×800)
Network-source policy	Ignore, Warn, Block

10.5 Update behavior

- Update check for VidiVerify itself on program start (can be enabled)
- Separate update check for the plug-ins on program start (can be enabled)
- A manual update check is always available from Settings

10.6 Support package

VidiVerify offers automatic assembly of a support package. It includes all relevant system information, protocols and diagnostic data, and can be sent to the support team on request. Assembly happens locally; sending it is up to you.

- Contains the current version, plug-in versions and runtime protocols
- Contains the anonymized device identifier, so support can match the package to your device unambiguously
- Contains no personal data or media content

10.7 Storage & maintenance

The Settings dialog is organized into four areas and includes the "Storage & maintenance" tab. There, VidiVerify automatically cleans up directories that grow over time, both on program start and at the click of a button — with individually adjustable scope.

- FFmpeg update backups: number of backups kept (default: the 2 most recent)
- Preview image cache: size limit in megabytes (default: 200 MB)
- Diagnostic dumps (hang dumps): number of files kept (default: 20)
- Update temp files: only deleted once older than five minutes

The "Clean up now" button lets you trigger cleanup manually at any time; a confirmation dialog shows the estimated space to be freed beforehand, and a completion dialog reports the result. Each area can also be set to "keep all" if desired.

> **[+] Principle: safe defaults**
> The application ships with conservative default settings suitable for most use cases. Changes are optional and can be reverted at any time.

11. File structure and storage layout

VidiVerify deliberately separates three areas: the actual program installation, user-specific data, and the machine-wide plug-ins. The application itself is installed in the program directory. Settings, protocols, the preview cache and optional statistics data reside in the personal user folder (%LocalAppData%). The large plug-in binaries (FFmpeg, MediaInfo, Magika) as well as the VidiVerify Runtime reside machine-wide under %ProgramData% — shared by every user of the machine. This keeps the installation tidy, makes runtime data transparently traceable, and allows updates or uninstalls to be carried out cleanly.

11.1 General description

Program folder

```
\VidiVerify
├── VidiVerify  (main program)
└── VidiVerify Uninstall
```

User folder (per user)

```
\VidiVerify
├── Settings and sources
├── Protocols and diagnostics
├── Preview cache and inbox
└── Optional statistics data
```

Plug-in folder (machine-wide)

```
\VidiVerify
└── tools (FFmpeg, MediaInfo, Magika, vv_runtime)
```

11.2 Example structure with file names

Program folder

```
C:\Program Files\VidiVerify
├── VidiVerify.exe
├── unins000.exe
└── unins000.dat
```

User folder (per user)

```
%LocalAppData%\VidiVerify
├── vidiverify_settings.json
├── module_sources.json
├── logs
│   ├── hangs
```

```
│    ├── exceptions.log
│    ├── faulthandler.log
│    ├── runtime.log
│    └── subprocesses.log
├── preview_cache
├── inbox
└── statistics
     ├── performance_history.jsonl
     └── performance_meta.json
```

Plug-in folder (machine-wide)

```
%ProgramData%\VidiVerify
└── tools
     ├── ffmpeg
     ├── mediainfo
     ├── magika
     └── vv_runtime
```

11.3 Notes on the key files

File or folder	Purpose
vidiverify_settings.json	Persistent user settings (language, paths, performance mode)
module_sources.json	Configurable sources for the plug-ins
logs/runtime.log	Chronological events from live operation
logs/exceptions.log	Exception errors with full stack traces
logs/subprocesses.log	Log of the plug-in processes started (FFmpeg, FFprobe, MediaInfo)
logs/faulthandler.log	Information on severe crashes
logs/hangs	Diagnostic dumps for unresponsive processes
preview_cache	Cached preview images (posters and slideshow thumbnails)
inbox	Handoff folder for files passed in from Explorer (Quick-Check / "Open with…")
statistics	Optional runtime and performance history
%ProgramData%\VidiVerify\tools\ffmpeg	Machine-wide FFmpeg and FFprobe binaries
%ProgramData%\VidiVerify\tools\mediainfo	MediaInfo plug-in for in-depth metadata analysis
%ProgramData%\VidiVerify\tools\magika	Magika plug-in for DeepCheck
%ProgramData%\VidiVerify\tools\vv_runtime	VidiVerify Runtime (Qt bundle with PyAV and OpenCV) for Media Lab and Media Test

[*] Transparency as a principle

All settings and protocols are stored in open, plain-text formats (JSON, text). This keeps them traceable outside the application too, and lets you evaluate or archive them however you need.

This storage layout improves maintainability, avoids unnecessary mixing of program files and user data, and results in a clearly traceable structure. Important settings and runtime data remain file-based, transparent, and easy to locate when needed. The application can be uninstalled cleanly via the bundled uninstaller.

12. Architecture and technical details

12.1 Overview

VidiVerify is developed in Python and uses the bundled standard library along with a selection of established third-party libraries. The main interface is built on Tkinter, Python's standard UI toolkit. The actual media analysis calls established, specialized command-line plug-ins (FFmpeg, FFprobe, MediaInfo, Magika). The more advanced, graphically demanding windows of Media Lab and Media Test, by contrast, run in their own, proprietary runtime environment — the VidiVerify Runtime, built on Qt. It is loaded automatically the first time it's needed, and comes bundled with a selection of high-quality Python media libraries (including PyAV and OpenCV). This keeps the lean main application small, while the compute-intensive analysis and rendering work runs in an environment built for it.

12.2 Component overview

Component	Role
User interface (Tkinter)	Input, display, operation and results presentation
Check module	Orchestrates the precheck, integrity check and full check
MAP (Media Analysis Protocol)	Proprietary evaluation and color coding of results
Plug-in manager	Manages the download, update and launch of the plug-ins
VidiVerify Runtime (Qt)	Its own, on-demand runtime environment for Media Lab and Media Test
Media libraries (PyAV, OpenCV)	Professional tools bundled in the runtime for decoding and image/frame analysis
Subprocess manager	Launches and monitors the plug-in processes
Logging system	Writes several levels of protocols to files in a structured way
Settings store	Persists user settings in JSON format
Watchdog	Monitors the user interface for unresponsive states
License module	Checks and unlocks the PRO edition
Support-package module	Automatic assembly of diagnostic information

[*] Professional tools under the hood

With the VidiVerify Runtime, VidiVerify brings a selection of professional media libraries along for the ride: PyAV binds the complete FFmpeg library for frame-accurate decoding, and OpenCV — the industry standard for computer vision — provides the foundation for the image and frame analyses (sharpness, motion,

> artifacts, continuity). VidiVerify orchestrates these high-quality tools deliberately, and translates their output into clear, understandable results.

12.3 Plug-in dependencies

- FFmpeg: decoding and checking of media files
- FFprobe: reading metadata and stream information
- MediaInfo: in-depth media analysis and supplementary metadata
- Magika: content-based file type detection for DeepCheck
- VidiVerify Runtime (Qt/PyQt-based): runtime environment for Media Lab and Media Test, bundling the following media libraries
- PyAV: Python binding to the FFmpeg libraries for frame-accurate decoding (in the runtime)
- OpenCV: image processing and frame analysis for the detail windows (in the runtime)
- PyQt and pyqtgraph: graphical rendering and interactive charts for the runtime windows
- psutil: detailed CPU and process information
- Pillow: image processing for the interface
- tkinterdnd2: drag-and-drop support
- fpdf: generation of the PDF protocols

12.4 Concurrency

The application uses several layers of concurrency to stay responsive. The user interface runs on the main thread, while checks run on separate worker threads. Communication between threads happens via a central message queue, which ensures that user-interface updates remain consistent.

12.5 Error handling

Errors are handled and logged at four levels:

Level	Handling
Operational notice	Informational message in the status bar and the protocol window
Warning	Visible highlight in the protocol, check can continue
Error	Check is aborted, a detailed protocol entry is written
Exception error	Full stack trace in exceptions.log, diagnostic dump if needed

12.6 Single-instance mechanism

The application prevents multiple instances from running at the same time, using a system-wide lock mechanism (mutex) plus a supporting lock file. If a second instance is started, a notice appears stating that the application is already running. Stale lock files are cleaned up automatically if the original process no longer exists.

12.7 Stable device identifier (VV-ID)

For support purposes, the application generates a stable, anonymized device identifier — the VV-ID (VidiVerify ID). It begins with the prefix VV- followed by five characters from a non-confusable character set. The identifier is based on a system-wide hardware identification plus a built-in salt value. It cannot be used to identify individuals or specific hardware.

12.8 Updates and integrity

Both the application itself and the plug-ins are sourced from signed or hash-verified locations. When a plug-in is downloaded, the archive's SHA-256 checksum is verified before installation. This reliably rejects tampered or damaged packages.

> **[*] Openness and verifiability**
> All formats, storage paths and protocol structures are designed to remain readable and analyzable even without the application. This keeps the user in full control.

13. Examples and use cases

13.1 Use case: intake inspection in a media archive

A media archive receives new video files every day from production teams and external suppliers. Before a file is admitted into the archive, it needs to be checked for structural integrity. The full check typically runs overnight, so daytime system load stays unaffected.

Typical setting: Full check, High-performance mode, strict mode on, unlimited time budget, exported as MAP+ (PRO) for audit-proof filing.

13.2 Use case: quick check before handoff

An editor wants to quickly check a finished episode before delivering it to the client. A full review isn't necessary, but typical container errors need to be ruled out. The integrity check delivers a solid interim result within a few minutes.

Typical setting: Integrity check, Balanced mode, MAP PDF for internal filing.

13.3 Use case: spot-checking in an archive audit

An archivist regularly checks samples from the collection to catch gradual media ageing. Every checked file receives a MAP+ PDF protocol, which is attached to the entry in the collection-management system.

Typical setting: Full check, Gentle mode during working hours, MAP+ as an archiving document (PRO).

13.4 Use case: forensic review

A forensic review calls for traceable checks with complete documentation. VidiVerify logs every check with runtime, timestamp, result and classification. The Protocol tab and the subprocess log are additionally available as a text-based chain of evidence.

Typical setting: Full check, strict mode on, maximum protocol limit, MAP+ as an archiving document.

13.5 Use case: media management with Kodi or Jellyfin

A private user or small operator wants to manage their collection with Kodi or Jellyfin. With the PRO edition, VidiVerify automatically generates an .nfo file that these systems read directly. The media library gets complete, technically accurate metadata with no manual upkeep.

Typical setting: Integrity check, NFO export enabled (PRO), automatic filing in the media folder.

13.6 Use case: routine IT maintenance

An IT department uses VidiVerify to regularly check media holdings stored on file servers. The goal is early detection of storage problems before entire generations of backups are affected. The update check for the plug-ins is kept active, so current decoders are always available.

Typical setting: Balanced mode, update checks enabled, Media Batch (PRO) for checking entire directories in series.

14. Planned extensions and outlook

VidiVerify is under continuous development. The following features are in preparation and form part of the strategic roadmap — mostly as PRO features, carrying the capability leap of the respective PRO tier. The order reflects current prioritization; final availability, scope and delivery date may change as development proceeds. Already-delivered features — including Media Lab with its compatibility rating, the Media Test detail windows for audio and video, DeepCheck, the Diagnostic tool, Quick-Check and the Spanish interface — are described in the preceding chapters.

14.1 AI forensics PRO

AI forensics assesses technical AI indicators in video files as evidence — strictly separated into an AI-likelihood score (how many characteristics suggest AI generation or editing) and a confidence score (how reliable that assessment is). The analysis runs entirely locally on the machine, with no cloud upload.

- An evidence-based assessment instead of a blanket yes/no verdict
- AI-likelihood score and confidence score reported separately
- Fully local processing, no upload of the file

14.2 "Media Repair" tab PRO

The "Media Repair" tab offers rule-based repair options for faulty video files. If a check finds a faulty file, VidiVerify suggests a suitable approach based on the specific error message and the repair options available in FFmpeg.

- A formatted error report for working through errors in detail
- Rule-based classification of errors with a probability rating
- A concrete repair suggestion with an explanation of the method
- Repairs are always applied to a copy, never to the original file
- The result is saved according to your choice

14.3 Batch processing of multiple files PRO

Batch processing lets you check multiple video files in a single run. A live results analysis is shown while processing.

- Processing entire directories or file lists
- A live results analysis while checking
- Automatic saving of the MAP reports per file
- A summary batch protocol with an overview

14.4 VVProgressiv™ — distributed, adaptive checking PRO

VVProgressiv™ checks short sequences distributed across the entire runtime, rather than just the start of the file — and automatically digs deeper wherever something looks suspicious. This produces fast confidence about the whole file, without immediately triggering a full decode. This is explicitly an intelligent sampling check, not a complete frame-by-frame check.

- Distributed check windows across the entire runtime, not just at the start
- Automatic deepening at anomalous points (adaptive escalation)
- Transparent reporting of the areas actually checked

14.5 MAP+ as an archiving PDF PRO

MAP+ is the even more extensive PDF variant of the MAP. It is designed as a professional archiving document and includes an expanded display with additional technical sections, and a structure optimized for long-term archive use. MAP+ is aimed at organizations, archives and professional users who pass check protocols on to clients or keep them for the long term.

14.6 Statistics module PRO

The statistics module builds an internal, learning statistics system within VidiVerify. It systematically records real check data and, over time, enables individual runtime forecasts.

- Systematic recording of real check data
- Long-term, system-specific runtime forecasts
- Detection of differences between systems, CPU modes and storage sources
- A solid data foundation for future analysis and UI features

14.7 Perceptual quality analysis (VQA) PRO

Perceptual quality analysis (VQA) extends the existing analysis features with a rating layer that reflects a video's visual perception from a human point of view. It answers the question: how good does the video actually look, independent of technical metrics?

- Visual quality assessment from the viewer's perspective
- A complement to technical metrics and platform ratings
- Uses OpenCV for the image analysis

14.8 YTUP™ — YouTube upload forecast PRO

YTUP™ is an integrated analysis module that assesses video files technically and qualitatively before a YouTube upload. The focus is deliberately not on a pure format check; it is a realistic forecast of how the video will look after YouTube's re-encoding.

- Technical assessment of upload suitability
- A quality forecast after YouTube's re-encoding
- Recommendations for optimizing before upload

14.9 macOS port

The application is planned to be ported to macOS as part of a refactoring effort. The goal is a consistent interface and comparable feature set across the supported operating systems.

14.10 Further languages

Further interface languages are planned:

- Hindi
- Russian
- Portuguese

14.11 NFO export for media management

In addition to the MAP, an .nfo file is planned for media management systems.

- Integration with systems such as Kodi, Jellyfin and Emby
- Uses existing technical analysis information
- A standards-compliant, robust and long-term-maintainable implementation

> **[*] Outlook**
> The extensions described here turn VidiVerify into a comprehensive platform for the quality control, analysis and management of video files. The modular architecture makes it possible to introduce these features step by step, without compromising the stability and simplicity of the existing application.

Appendix A: Glossary and terminology

The following overview explains key technical terms from the fields of video files, codecs, audio properties and the integrated plug-ins, insofar as they are relevant to VidiVerify. Terms are listed alphabetically.

Term	Explanation
AAC	Advanced Audio Coding, a lossy audio codec widely used in MP4 and AAC streams
AC3	An audio codec from Dolby, common on DVD and in broadcast
ALAC	Apple Lossless Audio Codec, lossless
Atmos	Object-based 3D audio format from Dolby, detected by MediaInfo since v1.4.7
AV1	A modern, royalty-free video codec with high efficiency, successor to VP9
AVC	Advanced Video Coding, identical to H.264
Bitrate	The amount of data per unit of time, typically in kilobits or megabits per second, often an indicator of a video or audio stream's quality
Bitrate EquivalenceLab	A tool in the Media Lab that converts bitrates between codecs: what bitrate in a more modern codec corresponds to a given bitrate in an older one, at comparable quality
Bit depth	The number of bits per audio sample or color channel; for audio, typically 16 or 24 bits — higher values mean finer resolution
Blu-ray	An optical storage medium for high-resolution video material, typically using MPEG transport streams
Broadcast	The television and radio distribution field, using specific container formats such as MXF and transport streams
Chroma subsampling	A technique for reducing color resolution; common variants are 4:4:4, 4:2:2 and 4:2:0
Codec	A rule set for encoding and decoding video or audio data, for example H.264, H.265, AAC
CodecLab	A tool in the Media Lab that classifies codecs and their properties, and helps choose the right format for a given purpose
Container	A file format that bundles video, audio and metadata streams into a single file, for example MP4, MKV, AVI
CRC	Checksum (Cyclic Redundancy Check), used to detect data errors
Decoding	Converting an encoded media file into displayable pictures and sound
DeepCheck	A deep type-integrity check using the Magika plug-in, a permanent part of the application since v1.4.7
Diagnostic tool	A three-tier deep analysis (streamcopy test, ffprobe deep analysis, decode layer) for stubborn cases, available since v1.4.7

Term	Explanation
DTS	Digital Theater Systems, an audio codec common on Blu-ray and DVD
DTS:X	Object-based 3D audio format from DTS, detected by MediaInfo since v1.4.7
EBML	Extensible Binary Meta Language, the structural foundation of the Matroska container format
Emby	A media management system that reads NFO files for metadata
Color space	A mathematical description of displayable colors; common spaces are BT.709 (HD) and BT.2020 (UHD)
FFmpeg	A plug-in for processing and decoding media files
FFprobe	A plug-in from the FFmpeg suite for reading metadata
FLAC	Free Lossless Audio Codec, a lossless audio codec
Frame	A single image within a video stream
Frame rate	The number of individual frames per second; typical values: 24, 25, 30, 50, 60 fps
H.264	A widely used video codec, also called AVC, the standard for HD video content
H.265	Successor to H.264, also called HEVC, more efficient at high resolutions
Hash value	A fixed string computed from a file, used for integrity checking, here SHA-256
HDR	High Dynamic Range, an extended dynamic range for bright and dark image areas
HEVC	High Efficiency Video Coding, identical to H.265
Integrity check	A check of file structure and decodability, based here on a 30-second sample
ISO BMFF	ISO Base Media File Format, the foundation for MP4 and MOV
Jellyfin	A free media management system that reads NFO files for metadata
JSON	A text-based data format for the structured storage of settings and metadata
AI forensics	A planned PRO module for an evidence-based assessment of technical AI indicators in video files, separated into an AI-likelihood score and a confidence score, running locally with no cloud upload
Classification	The assignment of a check result to a defined category, see Appendix B
Kodi	A widely used media management system that reads NFO files for metadata
Compatibility rating	A rule-based assessment of playability, based on technical media data
Performance mode	A predefined profile controlling CPU load during a check
Live telemetry	A display of system CPU, check-process CPU, RAM, network and read rate directly beneath the controls, available since v1.4.7
Magic bytes	A distinctive byte sequence at the start of a file that reveals the file format
Magika	A plug-in from Google for content-based file type detection

Term	Explanation
MAP	Media Analysis Protocol, VidiVerify's proprietary module for user-friendly presentation
MAP+	A high-quality PDF variant of the MAP as an archiving document, a PRO feature
MAP protocol	Export of the MAP as a designed, single-page A4 PDF archive sheet, via the "Save protocol" button in the main window
Matroska	An open, flexible container, also known as MKV
Media Lab	A dedicated tab serving as the launch pad for more advanced, detail-oriented analysis tools: the compatibility rating, plus the CodecLab, Bitrate EquivalenceLab and Video SizeLab tools
Media Repair	A planned tab for rule-based repair of faulty files, a PRO feature
Media Batch	A planned tab for batch processing, a PRO feature
Media Test	A multi-layered, technical detail analysis for audio and video, integrated into the Media Analysis tab (level, spectrum, bitrate, waveform, dropout and transcode detection)
MediaInfo	A plug-in for in-depth analysis of media files and stream parameters; has detected Atmos and DTS:X in main audio tracks since v1.4.7
Metadata	Descriptive information about a file, such as resolution, duration or codec
MKV	The file extension for the Matroska video container
Monkey's Audio (APE) / WavPack	Lossless compression formats
MOV	Apple's container format, technically related to MP4
MP3	A lossy audio codec, widely used
MP4	A widely used container format based on ISO BMFF
MPEG-2	A video compression standard, the foundation of DVD and digital television
MPEG-TS	MPEG transport stream, used in broadcast and on Blu-ray
MXF	Material Exchange Format, a professional container format for broadcast and production
NFO	A text-based metadata format for media management systems such as Kodi, Jellyfin and Emby
OpenCV	A library for image processing and computer vision, bundled in the VidiVerify Runtime
Opus	A modern, lossy audio codec with high quality
PCM	Pulse Code Modulation, uncompressed audio data
Perceptual quality analysis	An assessment of a video's visual perception, see VQA

Term	Explanation
Precheck	A brief, automatic check that runs when a file is loaded
PRO	A paid license tier for commercial users, with an expanded feature set
Protocol (log)	A chronologically ordered record of events during the application's operation
Quick-Check	A second-fast, preliminary check via right-click in Windows Explorer, using a lightweight worker component, without opening the full application
RIFF	Resource Interchange File Format, the structural foundation of the AVI container
Sample rate	The number of audio samples per second, measured in hertz (Hz); typical values: 44,100 Hz or 48,000 Hz
SHA-256	A cryptographic hash algorithm used for file integrity checking
Speex	A speech-optimized, lossy audio codec
Strict mode	A check mode in which the very first decoding error aborts the check
Stream	A single flow of data within a media file, usually separated by video, audio or subtitles
Subprocess	An external program launched and monitored by VidiVerify, for example a plug-in
Support package	An automatically assembled package of diagnostic information for the support team
Timeout	A time limit after which an operation is terminated
Transport stream	A continuous data stream with fixed packet sizes, the foundation for digital television
Video SizeLab	A tool in the Media Lab that estimates the effect of scaling, distinguishing plain upscaling (UPS) from AI-assisted methods (VSR)
VidiVerify Runtime	VidiVerify's own, on-demand Qt runtime environment with bundled professional libraries (PyAV, OpenCV) for Media Lab and Media Test
Full check	Complete decoding of the entire file for a thorough integrity check
Vorbis	An open, lossy audio codec used in the Ogg ecosystem
VP9	A video codec often used in the WebM container
VQA	Video Quality Assessment, the perceptual quality analysis
VV-ID (VidiVerify ID)	A stable, anonymized device identifier with the prefix VV-, used to match support cases; cannot be used to identify a person or specific hardware
VVProgressiv™	A planned PRO check that distributes short sequences across the entire runtime and automatically digs deeper at anomalous points
Watchdog	A monitoring component that watches the user interface for unresponsive states
WebM	An open container format based on Matroska, usually with VP9 or AV1
WMA	Windows Media Audio, both lossy and lossless variants

Term	Explanation
YTUP™	YouTube upload forecast, a planned PRO module from VidiVerify

Appendix B: Rating system and status classes

VidiVerify assigns every check a clear status class. This classification is part of the MAP (Media Analysis Protocol) and provides a quick read on the result. It forms the basis for follow-up decisions such as archiving, further processing, or re-copying.

B.1 Status classes

Status	Meaning
OK	The check completed without errors. The file is considered clean within the chosen check level.
WARNING	The check completed technically, but returned indications of minor anomalies. A manual assessment is recommended.
ERROR	The check encountered an error. The file shows structural or decoding problems and should not be used further without review.
NOT CHECKED	No result is yet available for this check level.

B.2 Video quality classes

In addition to the technical status class, the MAP also assigns the video content a quality class. The classification is based on resolution and provides a quick sense of the picture material's calibre.

Class	Typical resolution	Typical use
SD or smaller	up to 720 × 576	Older recordings, mobile footage from older devices
HD	1280 × 720	Standard web content, streaming, broadcast
Full HD	1920 × 1080	Home cinema, professional productions
2K to 4K	2048 × 1080 to 3840 × 2160	Cinema, high-quality streaming productions
Above 4K	more than 3840 × 2160	Specialty productions, planetariums, scientific use

B.3 Classification notes

In addition to the technical rating, certain notable characteristics are classified as text. Examples of such classifications include:

- Clean structure with no anomalies
- A clean check with no detectable errors
- Additional streams detected (for example subtitles, chapter markers)
- Timeout during the check

- Faulty file or stream access
- No streams detected
- Plug-in error in FFmpeg, FFprobe or MediaInfo

[i] The purpose of classification

The status class gives a quick, color-coded read on the result. The text classification adds the cause, so that even when there's an error, it's immediately clear where the problem comes from. Together, both are a core feature of the MAP.

Appendix C: Supported file formats

VidiVerify supports a broad range of common video and audio formats. Detection relies both on the file extension and on the file's header signature (magic bytes), so that even renamed files are classified reliably.

C.1 Container formats

Container family	Typical extensions	Description
MP4/MOV (ISO BMFF)	mp4, m4v, m4a, mov, 3gp, 3g2	Widely used — streaming, mobile devices, professional production
Matroska/WebM (EBML)	mkv, webm, mka	A flexible container, widely used on the internet
AVI (RIFF)	avi	A classic Windows container format
MPEG program stream	mpg, mpeg, vob, mod, tod	The DVD ecosystem, older consumer devices
MPEG transport stream	ts, m2ts, mts	Broadcast, Blu-ray, IPTV
Windows Media (ASF)	wmv, wma, asf	Older Windows-based productions
Flash Video	flv	Historical web content
MXF	mxf	Broadcast and professional production
Ogg	ogv, ogg	Open media formats
RealMedia	rm, rmvb	Historical streaming content

C.2 A note on codecs

The actual decoding is handled by the FFmpeg plug-in. All codecs available in the installed FFmpeg version are supported. These include, among others, H.264, H.265, VP8, VP9, AV1, MPEG-2, MPEG-4, AAC, MP3, FLAC, Opus, PCM and many more.

C.3 Audio file formats

As of version 1.4.9, VidiVerify also checks pure audio files as their own category (see section 6.12). The following audio file formats are detected and handled:

Audio family	Typical extensions	Description
MPEG audio	mp3, mp2	Widely used, lossy audio formats
MP4 audio (AAC)	m4a, aac	AAC audio in an ISO BMFF container, or as a raw stream

Audio family	Typical extensions	Description
FLAC	flac	Lossless compression at full quality
WAV / AIFF (PCM)	wav, wave, aif, aiff, aifc	Uncompressed PCM audio data at studio and archival quality
Ogg family	ogg, oga, opus, spx	Open formats with Vorbis, Opus or Speex audio
Windows Media Audio	wma	An audio format from the Windows Media ecosystem
Matroska audio	mka	An audio track in the flexible Matroska container
Lossless (APE / WavPack)	ape, wv	Monkey's Audio and WavPack, lossless compression
Dolby / DTS	ac3, eac3, dts	Multi-channel audio formats from Dolby and DTS, also as standalone files
AMR	amr	A speech-optimized format from mobile telephony

C.4 Audio codecs

The audio tracks of the formats listed above — as well as audio tracks within video containers — are decoded and analyzed using the following audio codecs:

Codec	Description
MP3 / MP2	MPEG audio, lossy, very widely used
AAC	Advanced Audio Coding, lossy, the standard in MP4
ALAC	Apple Lossless, lossless
FLAC	Free Lossless Audio Codec, lossless
PCM	Uncompressed audio data, as used in WAV and AIFF
Opus	A modern, lossy codec with high quality
Vorbis	An open, lossy codec in the Ogg ecosystem
Speex	A speech-optimized, lossy codec
WMA	Windows Media Audio, both lossy and lossless
Monkey's Audio (APE) / WavPack	Lossless compression formats
AC-3 / E-AC-3	Dolby Digital and Dolby Digital Plus, multi-channel audio
DTS	Digital Theater Systems, multi-channel audio, also DTS:X
AMR	Adaptive Multi-Rate, speech compression from mobile telephony

[+] Regular updates recommended

New codecs and container variants are continually added to FFmpeg. The automatic plug-in update check ensures that newer formats can be reliably recognized and checked as well.

Appendix D: Access points and controls

VidiVerify is operated entirely through the graphical interface. Classic keyboard shortcuts such as Ctrl+Shift+P are not currently available. The following overview summarizes the key access points and controls.

D.1 Main operation

Action	Access
Select a file	Browse button, or drag a file into the application area
Start a check	Start button in the main window
Cancel a check	Cancel button in the main window
Save the MAP protocol	"Save protocol" button after completion
Start the Diagnostic tool	"Start diagnostics?" button after a standard check reports an anomaly, or call it up from the Overview tab
Protocol tab	A tab in the main window with detailed runtime information
Open Settings	Settings button
Create a support package	An entry in the Settings dialog or the Help menu
Quit the application	Close the window; a confirmation prompt appears if a check is running

D.2 Error handling and diagnostics

- When exception errors occur, an entry is automatically written to exceptions.log
- When a process hangs, a diagnostic dump is saved to the logs/hangs folder
- The device identifier (Machine ID) is shown in the About dialog and can be shared with support requests
- The support package automatically assembles all relevant diagnostic information

[*] In closing

VidiVerify is designed as a reliable companion for everything related to video-file quality control. By combining a variety of established, high-performance and specialized plug-ins with the proprietary MAP, a clear interface and open data storage, VidiVerify becomes a tool that fits seamlessly into existing workflows and, with its planned extensions, keeps growing into a comprehensive platform for media quality checking.